



Namal University, Mianwali

Department of Computer Science

Machine Learning

CS-341 Machine Learning

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- Lecture breakdown and book is mentioned in the course outline (QOBE)
- Advanced level course with the following prerequisites
 - ▣ Probability Theory
 - ▣ Linear Algebra
 - ▣ Programming Skills (Matlab, Python)

CS-341 Machine Learning

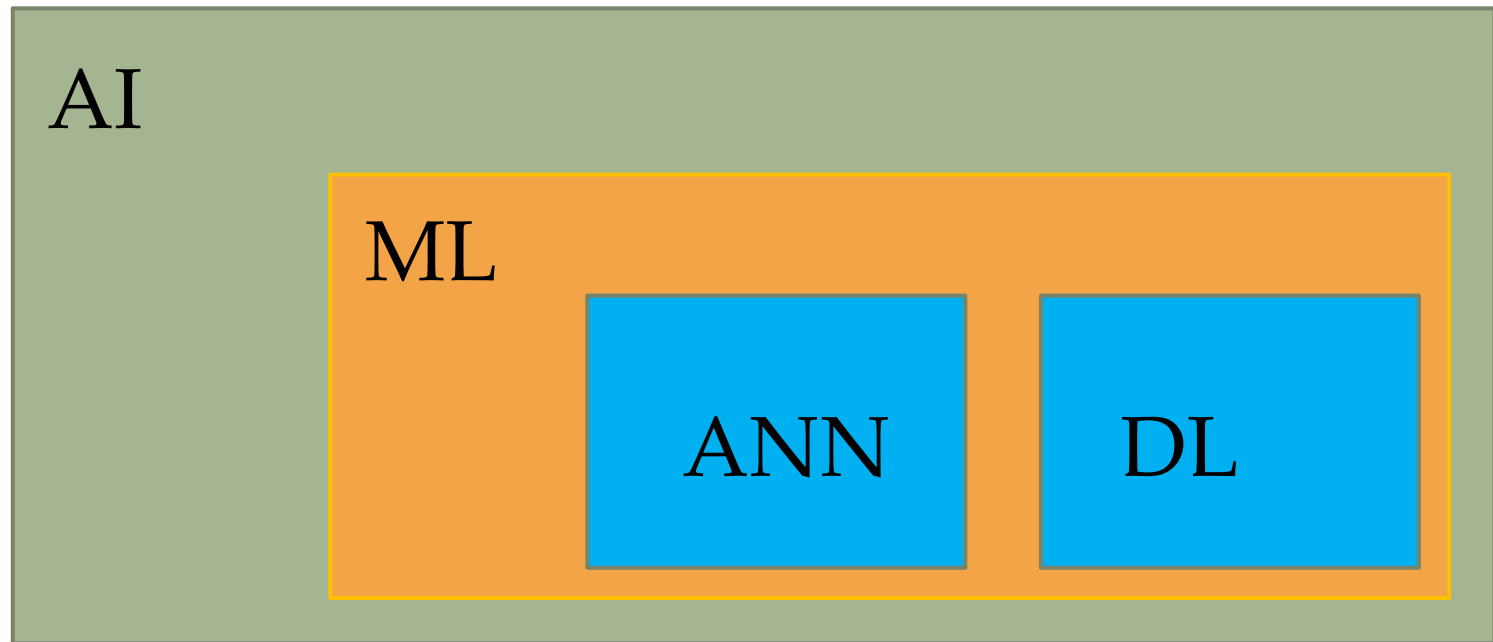
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□ Assessments and Evaluations

- | | |
|---------------|--------|
| □ Quizzes | 10% |
| □ Assignments | 10-15% |
| □ Project | 10-15% |
| □ Midterm | 25-30% |
| □ Final | 35-40% |

Machine Learning

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Machine Learning

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if x = this

y = that

else

y = not that

Given an observation

Guess the outcome

$$P(y|x) = \textit{that}$$

Machine Learning

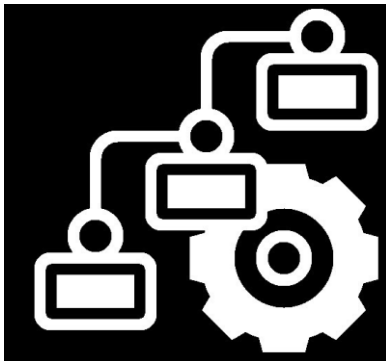
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if x = this

y = that

else

y = not that

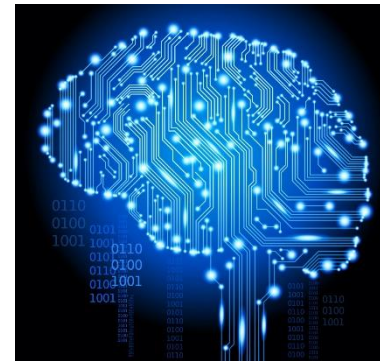


Not Intelligent

Given an observation

Guess the outcome

$P(y|x) = \text{that}$



Intelligent

Machine Learning

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if x = this

y = that

else

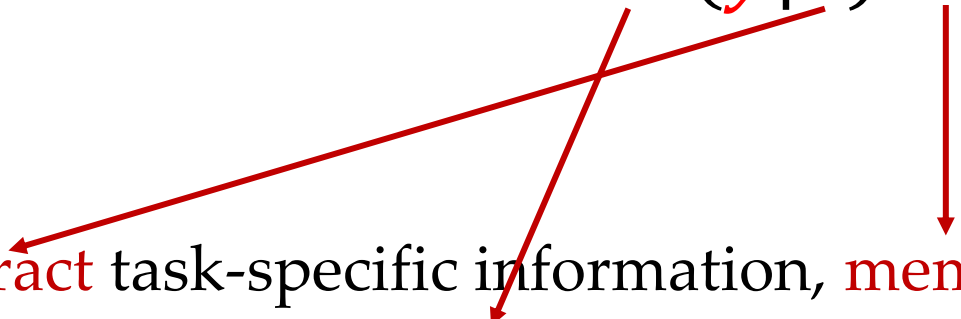
y = not that

Given an observation

Guess the outcome

$P(y|x) = \text{that}$

Learn to **extract** task-specific information, **memorize** and
able to make **estimated decision**

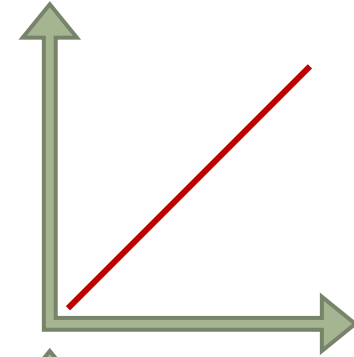


Why Machine Learning?

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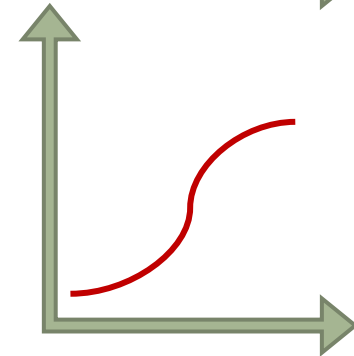
$$y = mx + c$$

Linear



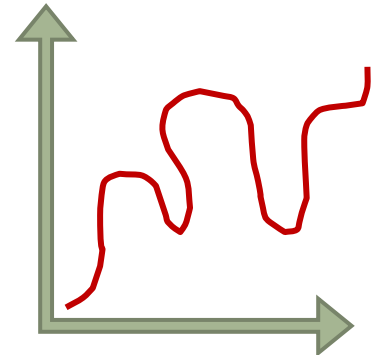
$$y = mx^3 + c$$

Nonlinear



$$y = (mx^5 + c)^9$$

Highly Nonlinear



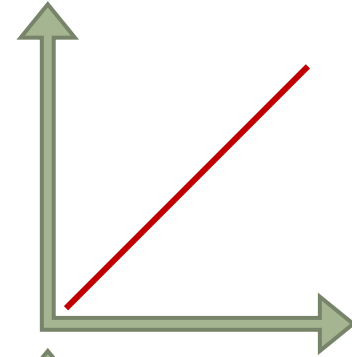
Why Machine Learning?

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$$y = mx + c$$

Linear

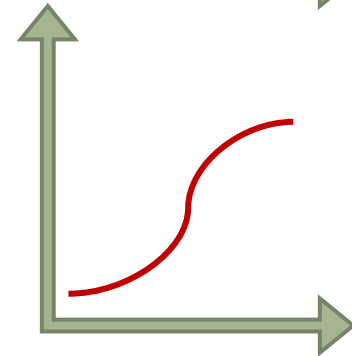
Direct Solution Possible



$$y = mx^3 + c$$

Nonlinear

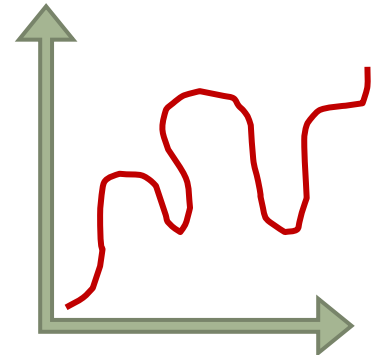
Direct Solution Possible



$$y = (mx^5 + c)^9$$

Highly Nonlinear

Direct Solution Not Possible --- ML



INTRODUCTION



Big Data

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- Widespread use of personal computers and wireless communication leads to “big data”
- We are both producers and consumers of data
- Data is not random, it has **structure**, e.g., customer behavior
- We need to extract that structure from data for
 - (a) Understanding the properties
 - (b) Learn and memorize
 - (c) Making predictions for the future

Why “Learn” ?

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- Machine learning is to program computers to optimize a performance criterion using sample data or past experience.
- Learning is used when:
 - ▣ Human expertise is not feasible
 - ▣ Humans are unable to explain their expertise (speech recognition)
 - ▣ Solution needs to be adapted to particular cases (user biometrics)

What We Talk About When We Talk About “Learning”

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- Learning **general models** from a data of particular examples
- Data is inexpensive and abundant (data warehouses, data marts); knowledge is expensive and scarce.
- Example in retail: Customer transactions to consumer behavior:
People who bought “X” also bought “Y”
- Build a model that is **a good and useful approximation** to the data.

Information Extraction

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- Retail: Market basket analysis, Customer relationship management (CRM)
- Finance: Credit scoring, fraud detection
- Manufacturing: Control, robotics, troubleshooting
- Medicine: Medical diagnosis
- Communications: Spam filters, intrusion detection
- ...

What is Machine Learning?

(to sum up)

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- **Extract** information from given **data**.
- Optimize a performance criterion using example data or past experience i.e., **think how to effectively and efficiently solve the problem**.
- Role of Statistics: **Statistical modeling** from sample data
- Role of Computer science: Efficient algorithms to
 - ▣ Solve the optimization problem
 - ▣ Representing and evaluating the models

Categories

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- Supervised Learning
 - ▣ Classification
 - ▣ Regression
- Unsupervised Learning
- Reinforcement Learning

Learning Associations

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□ Basket Analysis:

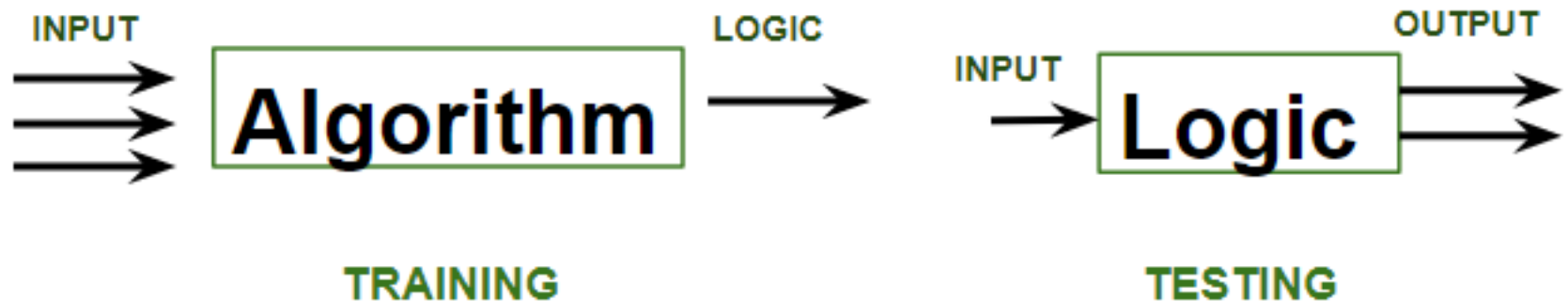
$P(Y | X)$ probability that somebody who buys X also buys Y where X and Y are products/services.

Example: $P(\text{chips} | \text{cola}) = 0.7$

Supervised Learning

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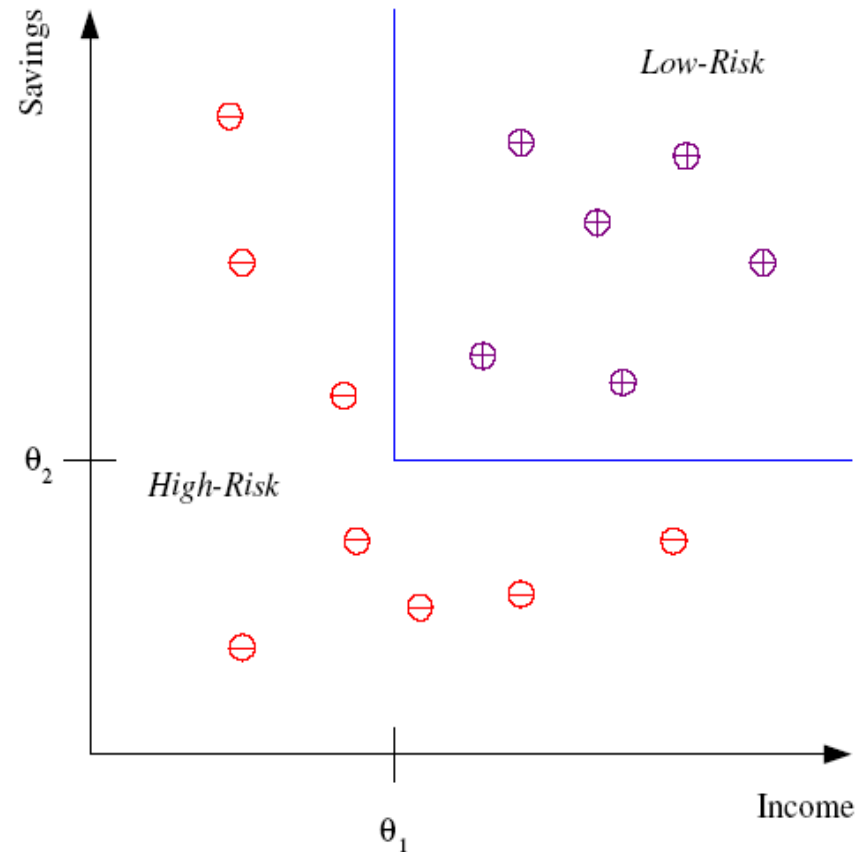
- Process of learning for machines that involve guidance or teaching to achieve a solution.
- For each data type, a **separate label** or identity is taught to the algorithm.



Classification

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- Example: Credit scoring
- Differentiating between **low-risk** and **high-risk** customers from their *income* and *savings*



Discriminant: IF *income* $> \theta_1$ AND *savings* $> \theta_2$
THEN **low-risk** ELSE **high-risk**

Classification: Applications

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- Aka Pattern recognition
- Face recognition: Pose, lighting, occlusion (glasses, beard), make-up, hair style
- Character recognition: Different handwriting styles.
- Speech recognition: Temporal dependency.
- Medical diagnosis: From symptoms to illnesses
- Biometrics: Recognition/authentication using physical and/or behavioral characteristics: Face, iris, signature, etc

Face Recognition

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Training examples of a person



Test images



ORL dataset,
AT&T Laboratories, Cambridge UK

Regression

□ Example: Price of a used car

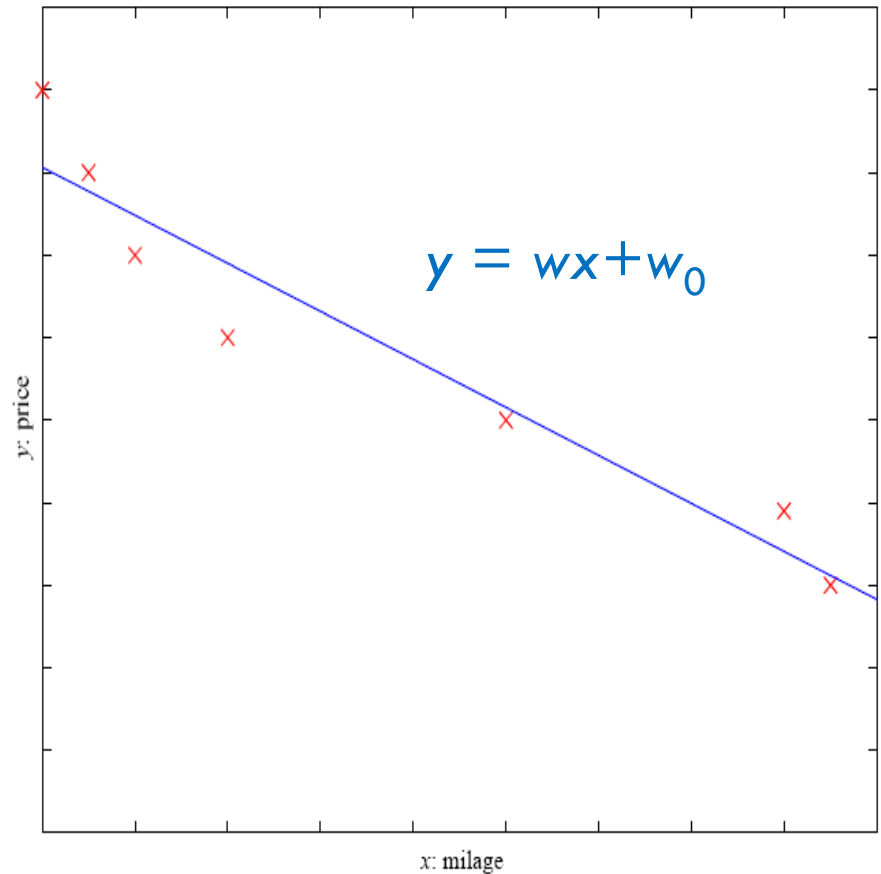
□ x : car attributes

y : price

$$y = g(x \mid \theta)$$

$g(\cdot)$ model,

θ parameters



Regression

□ Example: Price of a used car

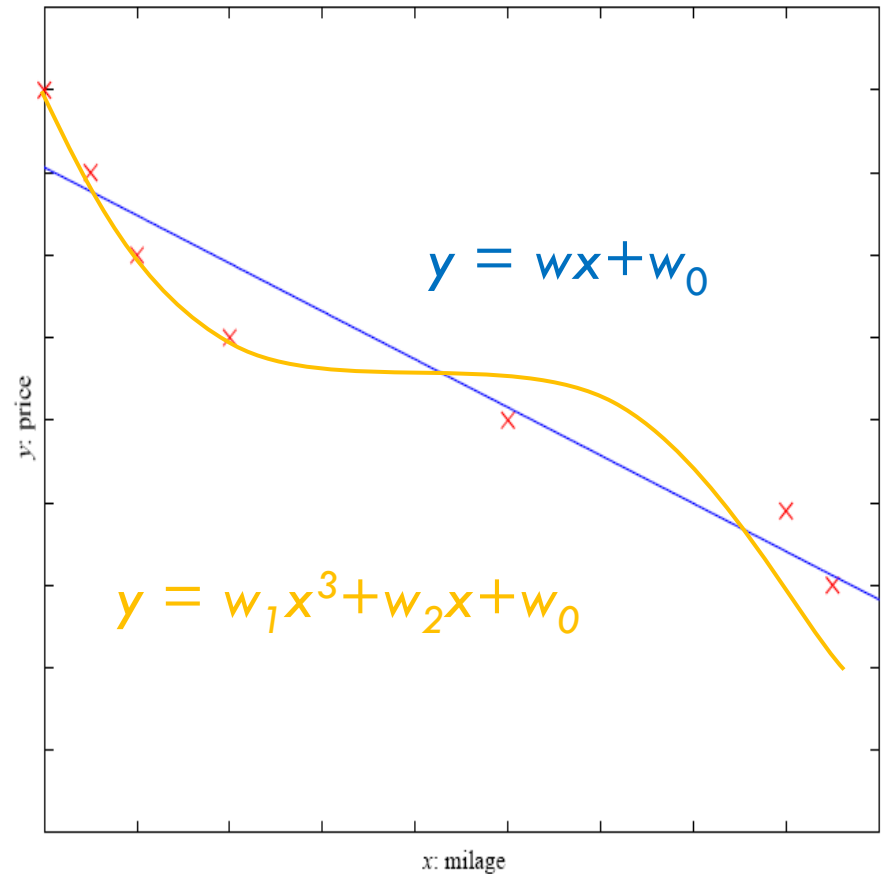
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Unsupervised Learning

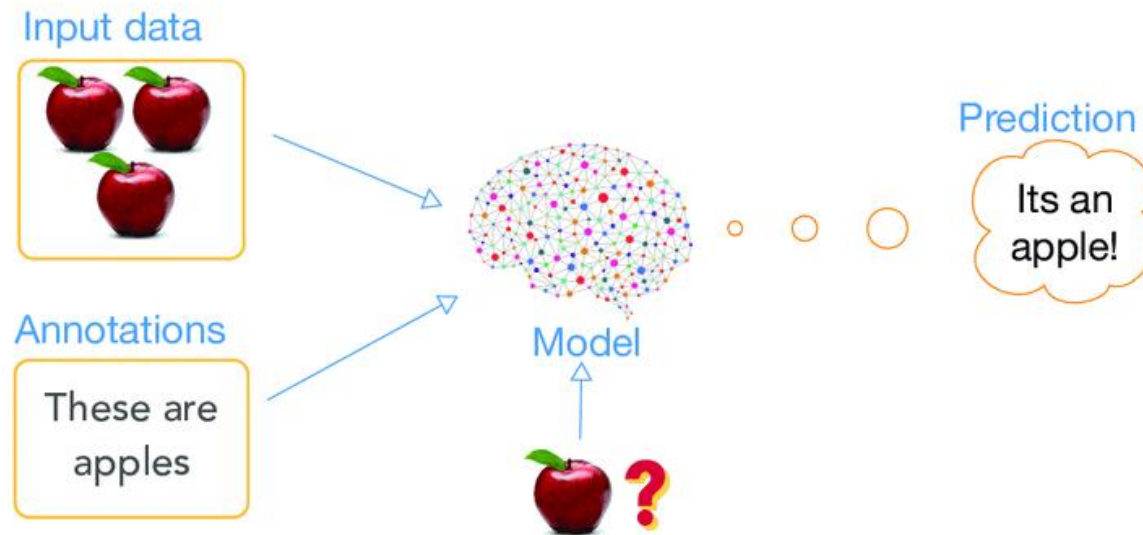
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- Learning “what normally happens”
- Clustering: Grouping similar instances using matching
- Example applications
 - ▣ Image compression: Color quantization
 - ▣ Quality assurance: Fruit sorting and grading

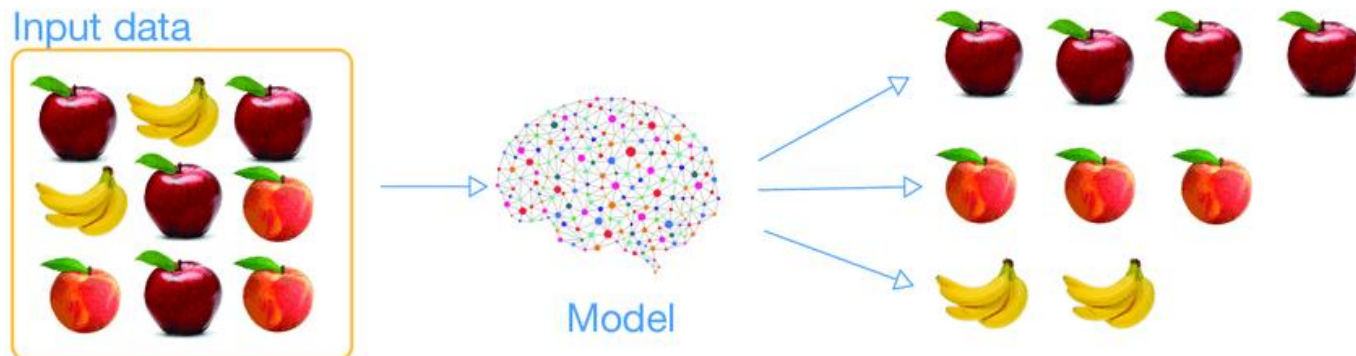
Supervised vs Unsupervised Learning

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supervised learning



unsupervised learning



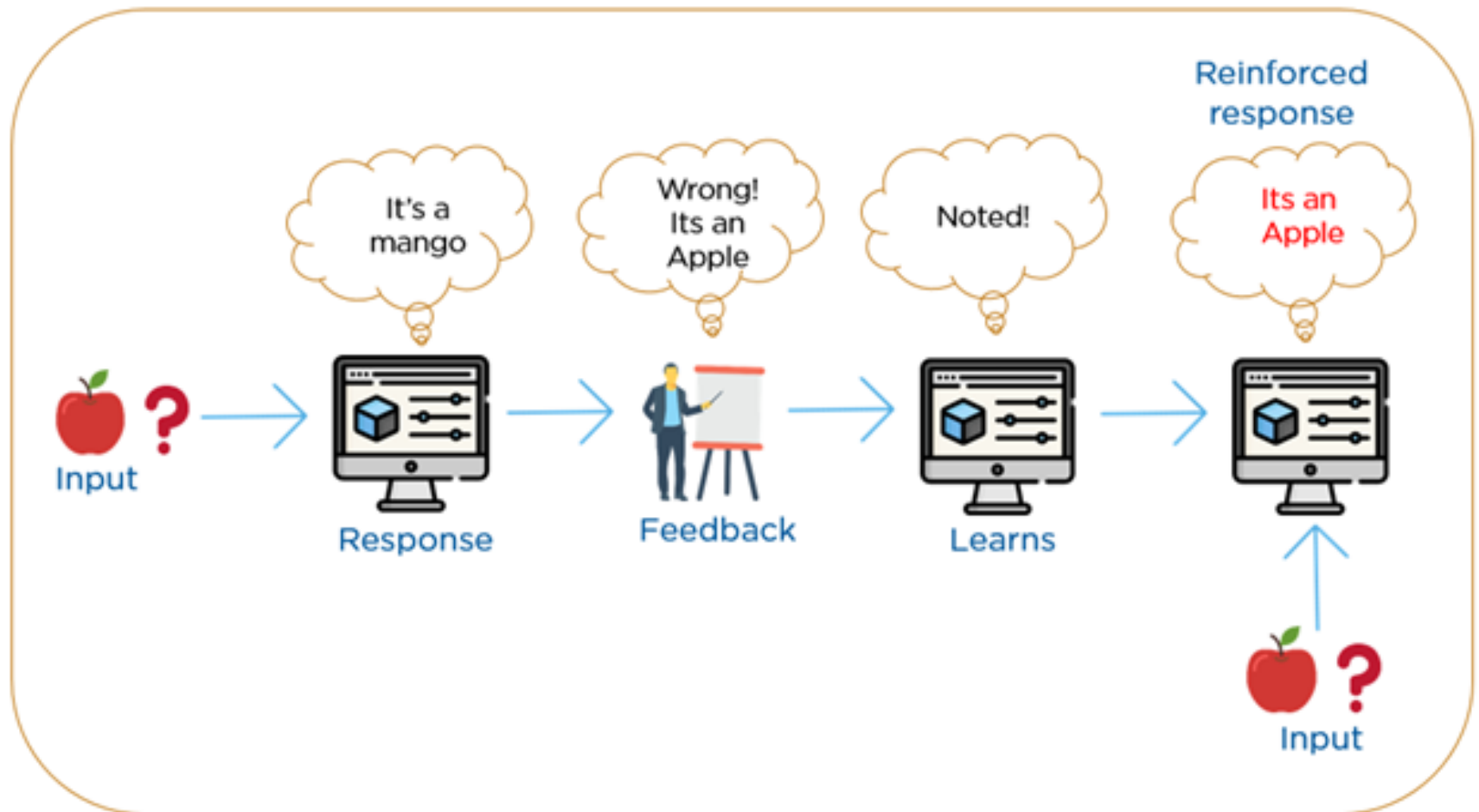
Reinforcement Learning

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- Learning a policy: A **sequence** of outputs
- No supervised output but delayed feedback
- Multiple agents, partial observability, ...
- **Learn from pure experience**
- Example: game playing

Reinforcement Learning

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A little test

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- Prediction about stock market
- Automatic seating plan according to age
- Speaker recognition
- Spam filtering
- Path finding robot
- Fruit sorting on conveyor belt
- Optical character recognition

A little test

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- Prediction about stock market = Supervised
- Automatic seating plan according to age = Unsupervised
- Speaker recognition = Supervised
- Spam filtering = Reinforcement
- Path finding robot = Reinforcement
- Fruit sorting on conveyor belt = Unsupervised
- Optical character recognition = Supervised

Thank You